

Diffused CO₂ Detection and Localizing the Subsea Leak Source with Robotic Fish

Muhammad Umar Masood¹, Qiang Zhu¹ and Zheng Chen²

Abstract

This paper presents a novel design and experimental testing of the robotic fish with three dimensionally maneuvering capability - AquaNavigator. The robotic fish has a capacity to carry one or more sensors for environmental safety and can monitor the sensing parameter in the underwater condition, such as Diffused Carbon Dioxide. Though localization in the depth of water by the means of Global Positioning Systems (GPS) is not possible, but this research presents the idea of collecting data underwater and tag it to the next possible surface position. For the purpose of testing and experiment, an indoor water tank is used to demonstrate the conditions, with a camera on the top to mimic the localization mechanism. A CO₂ pipeline leakage is setup in the bottom and the experiments show that the robotic fish is capable of localizing the area of leakage on CO₂. Further data analysis techniques are presented to accurately estimate the leakage source location. The results show that the leakage source can be estimated within an error of 0.69 m.

Index Terms

Bio-inspired robotic fish, CO₂ monitoring, underwater robotics, robotic fish design

I. INTRODUCTION

With the consumption of fossil fuels, the carbon dioxide (CO₂) levels in the atmosphere have been increasing rapidly, causing global warming and climate change [1], [2]. To mitigate these effects, carbon capture and storage (CCS) technologies have been developed to capture CO₂ emissions from power plants and industrial processes and store them underground and in offshore sequestration sites [3]–[5]. With the new policies and regulations around the United States and

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other countries, lower carbon footprint gives a competitive advantage to the companies. It is important for the regulatory bodies to ensure that this carbon is stored securely to the offshore sequestration sites, instead of being leaked and mixing in low pressure water bodies.

Leaked CO₂ from these underwater pipelines and sequestration sites have adverse effects on environment and marine ecosystems than the CO₂ already in air and not been captured [6], [7]. So, after the capture of CO₂, its safe transportation and storage is of crucial importance, both from environmental and economic perspectives. CO₂ is transported in a supercritical state through underwater pipelines to the sequestration sites. It is shown in literature that the impurities in CO₂, and the supercritical state induce phase transitions, leading to formation of a decompression plateau, which trigger long-distance crack propagation in the pipeline, resulting in severe consequences [8], [9]. Moreover, these pipelines are prone to leakages due to corrosion, material defects, and external damage. The leaked CO₂ can dissolve in water, leading to the formation of carbonic acid, which can lower the pH of the water and harm marine life [10], [11]. Therefore, it is essential to monitor the underwater pipelines for any potential CO₂ leakages and take necessary measures to prevent environmental damage.

There has been research efforts to inspect the subsea pipeline leakages using Autonomous Underwater Vehicles (AUVs) and Remotely Operated Vehicles (ROVs). Zhang et. al [12] presented an AUV which can detect the leakage of gas in the subsea pipelines using the visual aid, and detecting the gaseous bubbles in water. This does not specifically work for CO₂ leakages, but it is a valuable contribution towards the underwater pipeline leak detection. Qasem et. al [13] presented a multi-sensory ROV for oil spill surveillance. There are some efforts to detect leakage using the bio-inspired robotic fish as well. Hu et. al [14] developed a robotic fish to detect bubbles leakage in underwater pipelines. The authors used advanced Artificial Intelligence (AI) techniques to detect the bubbles in water using the on-board camera. All these systems are very technologically complex, expensive, and would still not provide a reliable solution for detecting CO₂ specifically.

Monitoring, preventive maintenance, and inspection of off-shore assets and ecosystem is a well established practice in the oil and gas industry for decades [15]–[17]. It is common to use Remotely Operated Vehicles (ROVs) and Autonomous Underwater Vehicles (AUVs) for such tasks, which require a dedicated team of expert operators and are expensive to operate. This cost is justifiable for maintenance operations and inspection of high-value assets. Moreover, these traditional ROVs do not fit to the oceanic ecosystem and can cause disturbance to the marine

life [18], [19]. On the other hand, a bio-inspired robotic fish can be a cost-effective solution which is more compatible with the marine ecosystem, and cause minimum disturbance.

In this presented work, a solution to the above problem is proposed by developing a novel bio-inspired robotic fish, AquaNavigator, which can be used to monitor the underwater CO₂ pipelines for any potential leakages. The robotic fish is designed to mimic the swimming behavior of real fish, which allows it to navigate through the water with minimal disturbance to the marine life. The robotic fish is equipped with a CO₂ sensor that can detect the presence of CO₂ in the water, keep record of the data, and tag it to the next possible surface position. The robotic fish can be deployed in the underwater environment to monitor the CO₂ pipelines for any potential leakages and provide real-time data to the operators.

For the purpose of this research, an indoor water tank is used to demonstrate the conditions, with a camera on the top to mimic the localization mechanism. A CO₂ pipeline leakage is setup in the bottom and the experiments show that the robotic fish is capable of localizing the area of leakage on CO₂. The experimental results demonstrate the effectiveness of the proposed solution in monitoring underwater CO₂ pipelines for potential leakages.

II. DESIGN AND DYNAMICS OF AQUANAVIGATOR

AquaNavigator is a novel bio-inspired robotic fish designed to mimic the swimming behavior of real fish. The robotic fish is designed to be compact, lightweight, and easy to maneuver in the water. The design of AquaNavigator is inspired by the morphology and swimming behavior of real fish, which allows it to navigate through the water with minimal disturbance to the marine life. The undulatory motion can be described by the traveling wave equation as introduced by Lighthill [20]:

$$y(x, t) = (c_1x + c_2x^2) \sin(kx + \omega t) \quad (1)$$

where y is the lateral displacement of the fish body at position x along the body length and time t , c_1 and c_2 are coefficients that define the amplitude envelope of the wave, k is the wave number, and ω is the angular frequency of the wave. This equation can be discretized, and thus for the case of a robotic fish, it needs at least two joints to mimic the undulatory motion, and it becomes:

$$y_i(x_i, t) = (c_{1i}x_i + c_{2i}x_i^2) \sin(k_ix_i + \omega t) \forall i = 1, 2 \quad (2)$$

To achieve this motion, the robotic fish tail uses a double slider-crank mechanism [21] which uses one actuator to generate this wave motion with a fixed phase difference between the two

joints, denoted as ϕ_1 . The detailed modeling and design of the robotic fish is presented in [21], which is an improved version of its squeal [22]. The depth control mechanism comes from [23]. AquaNavigator consists of a water tight enclosure for the circuitry, battery, buoyancy control mechanism and other prone-to-water components, a double slider-crank mechanism for tail fin actuation powered by a geared DC motor, and high-torque servo motor for the tail bias angle. Other than the water tight enclosure, all the components are individually water-proofed with marine compatible resin epoxy coatings. All the parts are 3D printed with SLA based resin printer, which provides high strength and durability. It is equipped with a three cell (3s) lithium-polymer (Li-Po) 2200 mAh battery pack for power and an ESP32-C3 microcontroller from Espressif Systems for control and communication. The robotic fish is also equipped with a Near Infrared (NIR) CO₂ sensor for detecting the diffused CO₂ levels in water. The detailed specifications of the robotic fish are presented in Table I and the annotated CAD model of the robotic fish is shown in Fig. 2. The fabricated real robotic fish is shown in Fig. 1.

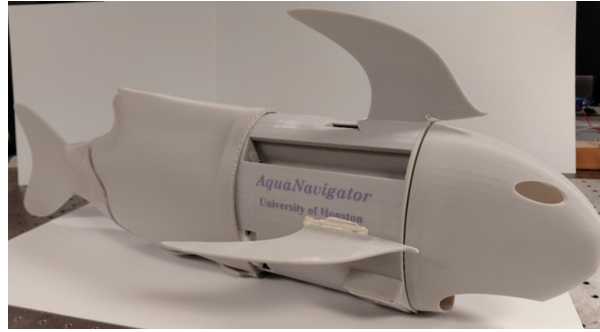


Fig. 1. Fabricated AquaNavigator robotic fish.

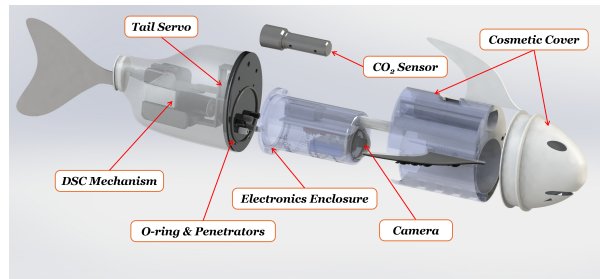


Fig. 2. Annotated CAD model of AquaNavigator robotic fish.

TABLE I
SPECIFICATIONS OF AQUANAVIGATOR ROBOTIC FISH

Parameter	Value	Unit
Speed	0.2	m/s
Speed	0.3	BL/s
Length	0.67	m
Swim Style	Carangiform	-
Weight	2.5	kg
Depth Rating	IP67	-
Depth Control Range	17.88	m
Battery Life	40	min
Operation Distance	480	m
On-board Sensors	NDIR CO ₂ sensor IMU Depth Camera	-

III. DIFFUSION OF CO₂

When CO₂ is leaked underwater, depending upon the source and shape of the leak point, its diffusion rate is affected by various factors. It is important to note, that any un-diffused CO₂ will rise to the surface and mix back with the atmospheric air, as where it was captured from. This CO₂ which mixed in air is not real concern, rather the diffused CO₂ is the one which can cause harm to the marine life. Under high static pressure of the water, the leaked CO₂ will dissolve in water, and a 3D cloud data of the diffused CO₂ can lead to the identification of the leak point. In this scenario, it was important for us to study the diffusion of CO₂ in water, so an experimental setup was established in an indoor water tank of 1.2m x 1.2m x 1.2m dimensions. Remember this tank is different than the tank used for the testing of the robotic fish. A CO₂ cylinder is connected to a regulator and a flow meter to control the flow rate of CO₂. The outlet of the flow meter is connected to a pipe which is submerged in water. The pipe has a small hole at the bottom to simulate the leak point. This experimental setup is shown in Fig. 3. Provided that the Near Infrared (NIR) CO₂ sensor has a response time of 60 seconds, before turning on the leakage, the based line reading of the sensor was noted. The CO₂ leakage was turned on at a flow rate of 1 liter per minute (LPM) and the readings were noted at different depths and different horizontal distances from the leak point. The CO₂ readings were noted after every 5

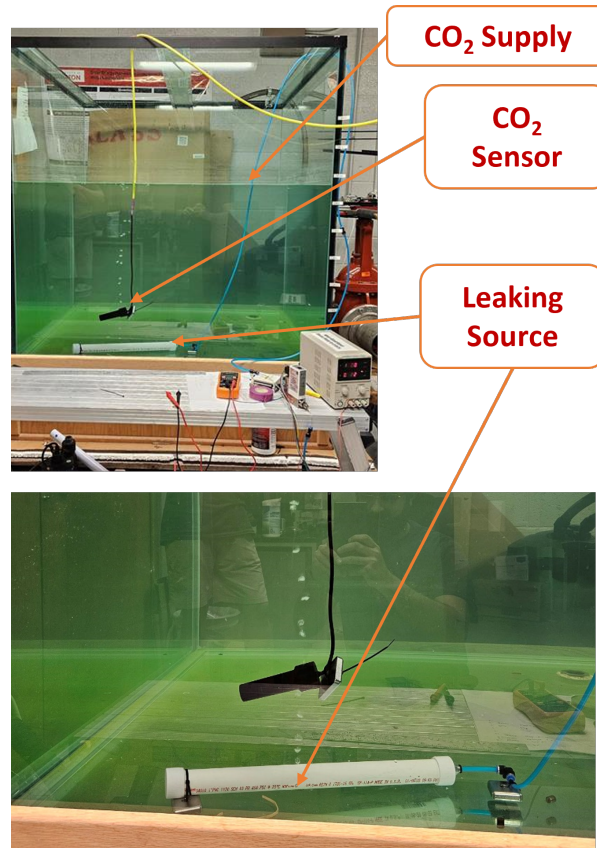


Fig. 3. Experimental setup for studying the diffusion of CO₂ in water.

minutes, to allow the sensor to stabilize. The CO₂ readings were noted at 3 different depths, 0.1m, 0.2m, 0.3m and 0.4m from the water surface, and at 5 different horizontal distances from the leak point, 0m, 0.1m, 0.2m and 0.3m. The CO₂ readings were noted for a total duration of 30 minutes. The experimental results are shown in Fig. 4. It can be observed that the CO₂ concentration is not highest near the leak point, rather it is highest at a certain distance from the leak point. This is due to the fact that the leaked CO₂ is in the form of bubbles, which rise to the surface and dissolve in water. As the bubbles rise, they dissolve in water and form carbonic acid, which lowers the pH of the water. The dissolved CO₂ then diffuses in water, leading to a higher concentration at a certain distance from the leak point. The experimental results also show that the CO₂ concentration decreases with increasing depth, as the static pressure of the water increases with depth, leading to a higher dissolution rate of CO₂ in water.

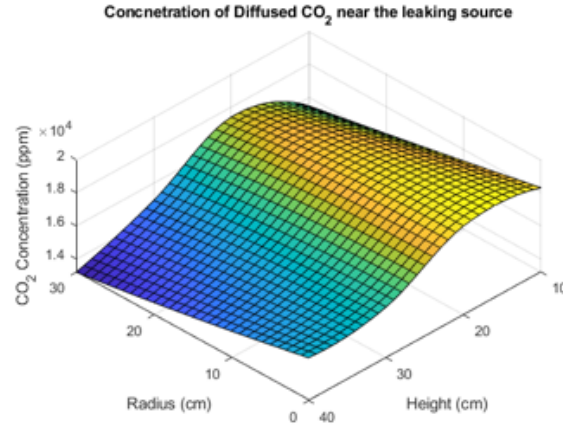


Fig. 4. Experimental results showing the diffusion of CO_2 in water at different depths and horizontal distances from the leak point.

IV. CO_2 MONITORING SOFTWARE AND DATA COLLECTION

A combined user-friendly software is developed to control the robotic fish and to monitor and record the sensed diffused CO_2 level in water. The software is developed in Python programming language and has a graphical user interface (GUI) to interact with the user. For the user's standpoint, the software has two main parts: (1) Manual Control of the robotic fish, and (2) Mission mode, which allows the user to set a predefined speed, depth rate, and curvature of the maneuvering path. With the mission mode, the robotic fish will move in a spiral path going downwards allowing it to scan a large area in water. The GUI of the software is shown in Fig. 5.

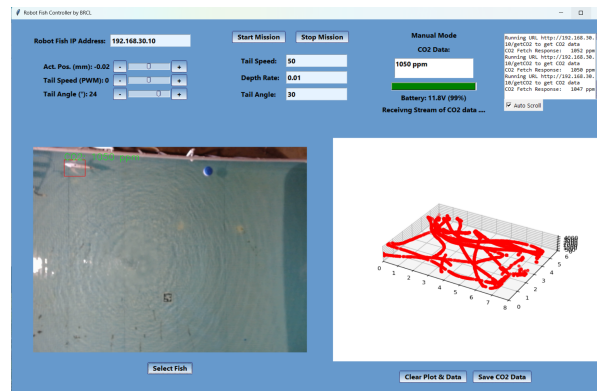


Fig. 5. Graphical User Interface (GUI) of the developed software for controlling the robotic fish and monitoring the CO_2 levels in water.

A. Experimental Setup

The experimental setup consists of an indoor water tank of dimensions 10 m x 5 m x 2 m, filled with about 70 percent of the height, with fresh water. An Red-Green-Blue (RGB) camera is mounted on the top of the water tank, which covers about 8 m x 6 m area of the water tank. The camera is connected to the computer and the software uses computer vision techniques to track the position of the robotic fish in the water tank. A CO_2 cylinder is connected to a regulator and a fixed leak rate was set from the tank. The outlet of the regulator is connected to a 2 inch diameter pipe which is submerged in water and has a small hole at the bottom to simulate the leak point as shown in Fig. 6. The CO_2 would come of the leakage holes in the form of bubbles, rise to the surface and dissolve in air. As evident from the lab experiment in the previous section, some of this CO_2 will dissolve in water and diffuse in water. The ratio of CO_2 which dissolves in water depends upon the static pressure of the water, flow rate of the leakage, temperature of the water and other factors. The purpose of this study to use the data analysis techniques to localize the leak point without considering all those complex parameters.

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The robotic fish is placed on the surface of the water, and been controlled manually from the software. The initial detection of the robotic fish is done by the human assistance, and then the visual tracker keeps the postion of the robotic fish using optical flow techniques. Through out the manual operation of the robotic fish, the diffused CO_2 levels are sensed and being sent



Fig. 6. CO₂ leakage source setup in the water tank.

to the host computer and been live plotted in the software. The complete experimental setup is shown in Fig. 7.

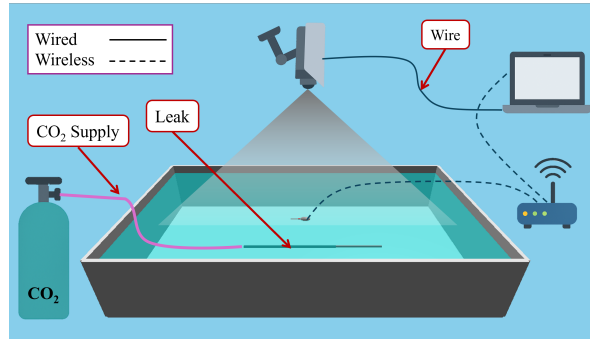


Fig. 7. Complete experimental setup for testing the robotic fish and monitoring the diffused CO₂ levels in water.

B. Communication

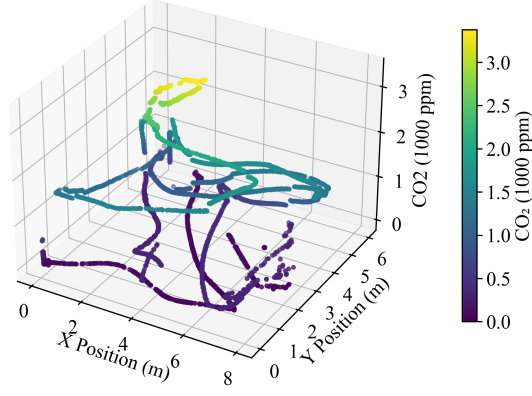
The robotic fish is equipped by a ESP32-C3 microcontroller, which has built-in wireless IEEE 802.11 (Wi-Fi). For the demonstration purposes, it is good enough to communicate with this wireless in an indoor water tank. However, for the real-world applications, a more robust communication system is required, such as very low frequency (VLF) radio waves or acoustic communication. The idea remains the same, when the robotic fish is not able to successfully send the data through wireless communication, it will store the data in its internal memory and send it when it is back in the range of communication. At the minimum, the robotic fish will be

able to send the data when it surfaces. Other than communication frequency, the data sending mechanism will remain the same.

The robotic fish and the computer communicate over a TCP/IP socket connection. The robotic fish acts as a server and the computer acts as a client. The computer sends control commands to the robotic fish, and the robotic fish sends back the status and sensor data. The communication protocol is simple and easy to implement, making it suitable for real-time applications. The socket requests are sent as scheduled tasks for different functionalities, such as sending control commands, receiving status updates, and receiving sensor data. When the computer software does not receive any response from the robotic fish within a certain time, it assumes that the connection is lost and intimates the user. If the robotic fish does not receive any command from the computer within a certain time, it assumes that the connection is lost, keeps storing the sensor data, and stops moving to avoid any accidents, unless it is in mission mode. Fig. 7 shows the communication flow between the robotic fish and the computer.

C. Geo tagging of CO₂ Data

We know the limitation of the Global Positioning System (GPS) in an indoor environment, and it is not practical to use GPS in indoors. So, for the purpose of this research, a camera is mounted on the top of the water tank to provide the localization of the robotic fish. The camera is connected to the computer and the software uses computer vision techniques to track the position of the robotic fish in the water tank. The camera captures the images of the water tank and processes them to detect the robotic fish. The position of the robotic fish is then calculated based on its location in the image. The software uses OpenCV library for image processing and computer vision tasks. The position of the robotic fish is then used to tag the CO₂ data with its corresponding location in the water tank. Irrespective of the mission mode, the software will always keep plotting the diffused CO₂ levels corresponding to its position in the 3D plot in the GUI. The 3D plot is updated in real-time as the robotic fish moves in the water tank. The user can also save the CO₂ data with its corresponding location in a CSV file for further analysis. The 3D plot of the diffused CO₂ levels in water tank is shown in Fig. 8. The data analysis of this data is further discussed in the next section.

Raw CO₂ Samples: 3D Point Cloud (no temporal aggregation)Fig. 8. 3D plot showing the diffused CO₂ levels in water tank. The color bar shows the CO₂ concentration in ppm.

V. RESULTS AND DISCUSSION

The data of diffused CO₂ levels in the water tank is collected by the robotic fish and tagged with its corresponding position in the tank. Although the data is collected over time, with each data point being approximately 0.1 seconds apart, the processing of the full spatiotemporal dataset is computationally expensive. Furthermore, considering that the time response of state-of-the-art dissolved CO₂ sensors is on the order of 60 seconds, it is reasonable to assume that the data collected over a short period of time can be treated as quasi-steady. Hence, for the purpose of the present analysis, the data is regarded as spatial, and the temporal component is ignored. This assumption allows the use of spatial interpolation and binning techniques to visualize and analyze the CO₂ diffusion pattern. If the leakage source can be localized without explicitly modeling temporal evolution, the approach provides an efficient means to identify the leak origin.

A. Spatial Data Analysis

To obtain a spatially continuous representation of the CO₂ field, the raw measurements are discretized into a uniform 2D grid, and the average CO₂ concentration within each cell is computed to generate a binned heatmap. The mean concentration $\bar{C}(x_i, y_j)$ in each grid cell

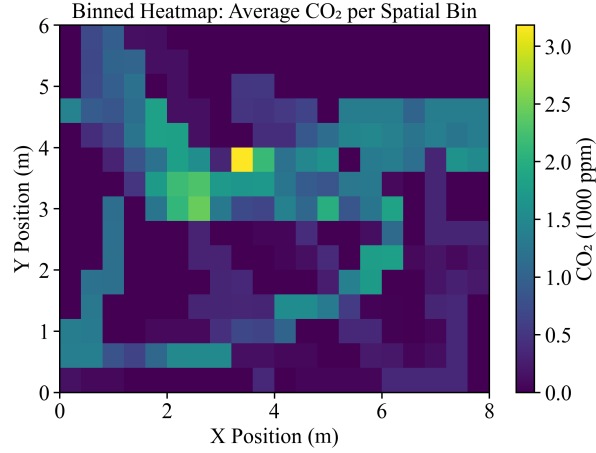


Fig. 9. 2D heatmap showing the diffused CO₂ levels in water tank, with a grid size of 0.4m x 0.4m.

centered at (x_i, y_j) is given by

$$\bar{C}(x_i, y_j) = \frac{\sum_{n=1}^N C_n \delta_{ij}(x_n, y_n)}{\sum_{n=1}^N \delta_{ij}(x_n, y_n)} \quad (3)$$

$$\delta_{ij}(x_n, y_n) = \begin{cases} 1, & x_n \in [x_i - \frac{\Delta x}{2}, x_i + \frac{\Delta x}{2}), \\ & y_n \in [y_j - \frac{\Delta y}{2}, y_j + \frac{\Delta y}{2}), \\ 0, & \text{otherwise.} \end{cases}$$

Here, C_n denotes the CO₂ concentration measured at location (x_n, y_n) , and $(\Delta x, \Delta y)$ are the grid spacings in the x and y directions, respectively. The indicator function δ_{ij} assigns each measurement to the corresponding grid cell based on its spatial coordinates. The resulting two-dimensional heatmap, shown in Fig. 9, captures the spatial distribution of CO₂ concentration within the water tank. It can be observed that the CO₂ concentration is highest near the leak point, which is expected due to the localized source. Other regions with elevated concentrations may correspond to zones of CO₂ diffusion and mixing within the water tank. This method is the computationally most effective, but may not be very accurate about estimating the leak point. As seen in the figure, the leak point is detected near (3.5, 4). The scattered CO₂ measurements $\{(x_n, y_n, C_n)\}_{n=1}^N$ were interpolated onto a regular grid (x_i, y_j) using piecewise linear interpo-

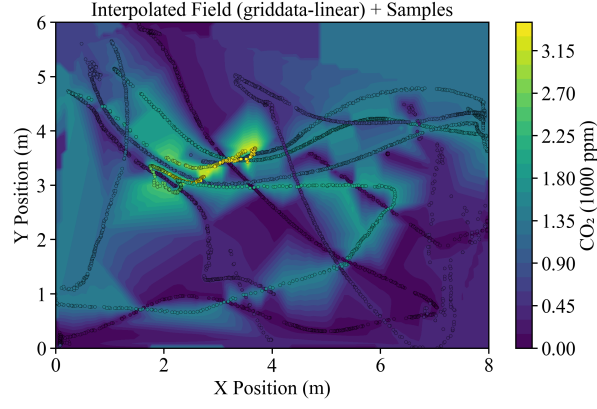


Fig. 10. Linear and nearest interpolation of the diffused CO₂ levels in water tank

lation. The interpolated field $C_g(x_i, y_j)$ is given by

$$C_g(x_i, y_j) = \sum_{n=1}^N \lambda_n(x_i, y_j) C_n, \quad \text{with} \quad \sum_{n=1}^N \lambda_n(x_i, y_j) = 1, \quad (4)$$

where the interpolation weights $\lambda_n(x_i, y_j)$ are the barycentric coordinates of the point (x_i, y_j) with respect to the Delaunay triangle enclosing it. For grid points lying outside the convex hull of the sampled positions, the nearest-neighbor interpolation is used to assign a value:

$$C_g^{\text{filled}}(x_i, y_j) = \begin{cases} C_g(x_i, y_j), & (x_i, y_j) \in \Omega_{\text{hull}}, \\ C_{n^*}, & (x_i, y_j) \notin \Omega_{\text{hull}}, \end{cases} \quad (5)$$

where Ω_{hull} denotes the convex hull of the measurement locations and n^* is the index of the nearest sample point. More for visual purposes, and less for direct leak localization, a radial basis function (RBF) interpolation is also applied to the scattered CO₂ measurements. The interpolated field $C_r(x_i, y_j)$ is expressed as

$$C_r(x_i, y_j) = \sum_{n=1}^N w_n \phi(\|(x_i, y_j) - (x_n, y_n)\|), \quad (6)$$

where $\phi(r)$ is a radial basis function, and the coefficients w_n are determined by enforcing the interpolation condition $C_r(x_n, y_n) = C_n$ for all $n = 1, \dots, N$. A commonly used choice for $\phi(r)$ is the Gaussian kernel:

$$\phi(r) = \exp\left[-\left(\frac{r}{\epsilon}\right)^2\right], \quad (7)$$

where ϵ is a shape parameter controlling the smoothness of the interpolation. The resulting RBF-interpolated field, shown in Fig. 11, provides a smooth and visually continuous representation of

RBF Interpolation: Smooth Spatial Surface

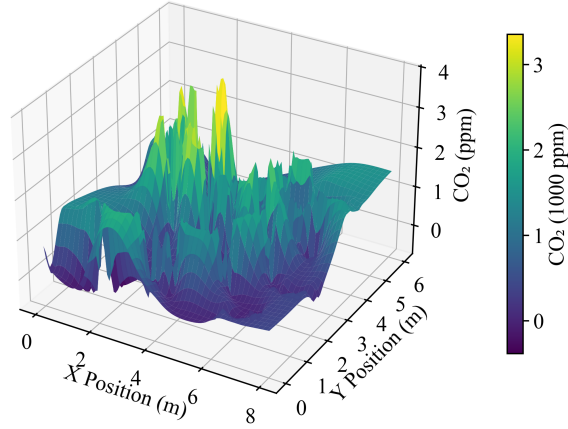


Fig. 11. Radial basis function interpolation of the diffused CO_2 , the peaks show the possible leak points.

the CO_2 concentration across the water tank, highlighting diffusion patterns while suppressing small-scale measurement noise.

B. Localizing the Leakage Sources

To identify potential leak points from the interpolated CO_2 field, a peak detection algorithm is applied to locate local maxima in the concentration distribution. The algorithm scans the interpolated grid and identifies points that are higher than their immediate neighbors, indicating a local peak. To reduce false positives due to noise, a minimum prominence threshold is set, ensuring that only significant peaks are considered. The detected peaks are then ranked based on their concentration values, with the highest peaks being the most likely candidates for leak sources. The highest three peaks are averaged to estimate the most probable leak location, which is marked on the plot in Fig. 12. This method provides a straightforward and computationally efficient approach to localizing leak points based on spatial concentration patterns, although further validation with ground truth data would be necessary for practical applications. Figure 12 presents the detected local maxima (red crosses) in the RBF-interpolated CO_2 concentration field. The top three peaks (red stars) are selected as the most likely leakage candidates, and their averaged coordinates (cyan marker) estimate the final leak position. The estimated leak position lies close to the actual source location at (3.0, 3.5) m, demonstrating the effectiveness of the concentration-based spatial peak detection approach.

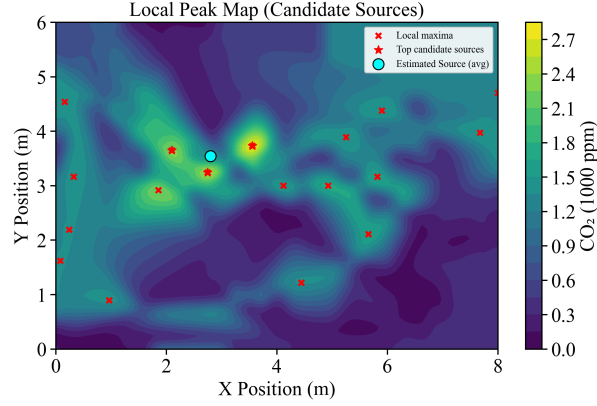


Fig. 12. Peak map showing the possible leak points in water tank, based on the local maxima points and averaging the highest three points.

C. Real Time Detection

As described in the previous sections, the robotic fish is maneuvering arbitrarily in the suspected area of the CO_2 leakage, and collecting the data of diffused CO_2 levels in water, with geo-tag and timestamps of the data, a feasibility is performed to see how accurately and effeciently, the same data can be used in real time for the robot to reach to the leak point. Considering the computational time limitation, the robot can process data after specified time. And each now data analysis provides the robots a better and better estimation of the leak source. For this purpose, the data is divided into multiple segments, each segment contains data collected over a period of 75 seconds. Each segment is processed using the same data analysis techniques as described in the previous sections, and the estimated leak point is recorded after each segment.

Fig. 13 shows a series of spatial CO_2 concentration maps generated at different time intervals, demonstrating the progressive refinement of leak localization as more data is collected. The figure displays nine snapshots taken at 75s, 151s, 226s, 301s, 376s, 452s, 527s, 602s, and 678s, respectively. Each subplot presents the RBF-interpolated concentration field with color gradients indicating CO_2 levels in ppm, where yellow regions represent high concentrations and purple regions indicate low concentrations. The actual leak source is marked with a pink diamond at coordinates (3.0, 3.5) m, while the estimated source position from the algorithm is shown as a cyan circle. Local maxima detected in the concentration field are indicated by orange crosses, and the top three candidate peaks are marked with red stars.

The localization error, displayed in the upper left corner of each subplot, demonstrates signifi-

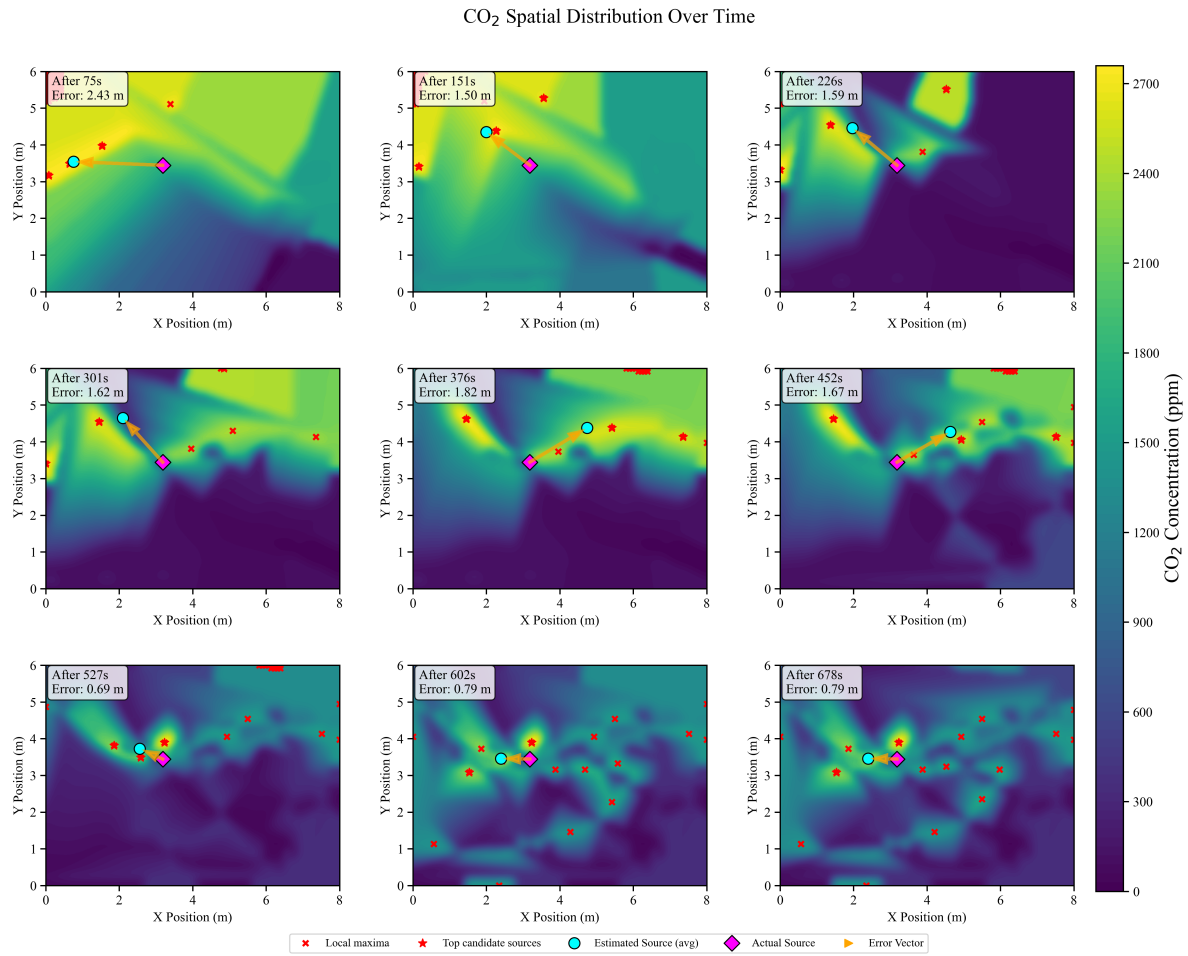


Fig. 13. CO₂ spatial distribution over time showing progressive improvement in leak localization. Each subplot represents a cumulative data window of 75 seconds, with the estimated source (cyan circle) converging toward the actual source (pink diamond) as localization error decreases from 2.43 m to 0.69 m.

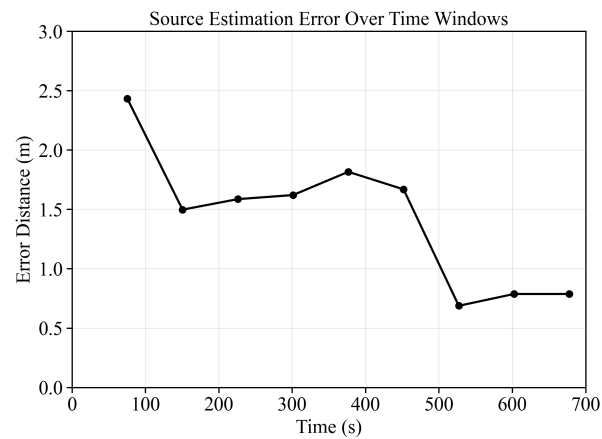


Fig. 14. Leak localization error over time, showing a decreasing trend as more data is collected.

cant improvement over time. Starting with an initial error of 2.43 m after 75 s of data collection, the error progressively decreases to 1.50 m at 151 s, and continues to reduce through subsequent time windows. By 527 s, the error reaches 0.69 m, showing convergence toward the true leak location. This temporal progression illustrates the effectiveness of the incremental data processing approach, where each additional 75-second segment contributes to a more accurate spatial representation of the CO₂ distribution. The concentration maps become increasingly refined and localized around the actual source, validating the feasibility of real-time leak detection using the proposed methodology. The results demonstrate that even with limited initial data, the robotic fish can provide useful estimates that improve continuously as it collects more measurements, enabling adaptive navigation strategies for efficient leak localization. Fig. 14 illustrates the trend of leak localization error over time, showing a clear decrease as more data is collected. The plot indicates that the error reduces from an initial value of 2.43 m to 0.69 m after 527 s, highlighting the effectiveness of the data-driven approach in refining leak source estimates with increasing measurement duration.

VI. CONCLUSION

A bio-inspired robotic fish, AquaNavigator, is designed and developed to monitor the underwater CO₂ pipelines for any potential leakages. The robotic fish is equipped with a CO₂ sensor that can detect the presence of CO₂ in the water, keep record of the data, and tag it to the next possible surface position. The robotic fish can be deployed in the underwater environment to monitor the CO₂ pipelines for any potential leakages and provide real-time data to the operators. An indoor water tank is used to demonstrate the conditions, with a camera on the top to mimic the localization mechanism. A CO₂ pipeline leakage is setup in the bottom and the experiments show that the robotic fish is capable of localizing the area of leakage on CO₂. The experimental results successfully demonstrate that the proposed technique can be used to localize the leakage source of CO₂ in underwater pipelines, without using a swarm or multiple sensing nodes to collect a combination of the spatial and temporal data. The results show that the spatial data alone, when collected for a sufficiently long time, can be used to localize the leak point. The proposed technique is computationally efficient and can be used in real-time applications. Future work will focus on improving the accuracy of the leak localization by incorporating temporal data and using more advanced interpolation techniques.

REFERENCES

- [1] M. Filonchik, M. P. Peterson, L. Zhang, V. Hurynovich, and Y. He, "Greenhouse gases emissions and global climate change: Examining the influence of co₂, ch₄, and n₂o," *Science of The Total Environment*, vol. 935, p. 173359, 7 2024. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S004896972403506X>
- [2] M. Kabir, U. E. Habiba, W. Khan, A. Shah, S. Rahim, P. R. los Rios-Escalante, Z. U. R. Farooqi, and L. Ali, "Climate change due to increasing concentration of carbon dioxide and its impacts on environment in 21st century; a mini review," *Journal of King Saud University - Science*, vol. 35, p. 102693, 7 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1018364723001556>
- [3] H. J. Herzog, "What future for carbon capture and sequestration?" *Environmental Science & Technology*, vol. 35, pp. 148A–153A, 4 2001.
- [4] S. Yadav and S. S. Mondal, "A review on the progress and prospects of oxy-fuel carbon capture and sequestration (ccs) technology," *Fuel*, vol. 308, p. 122057, 1 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0016236121019335>
- [5] A. I. Osman, M. Hefny, M. I. A. Maksoud, A. M. Elgarahy, and D. W. Rooney, "Recent advances in carbon capture storage and utilisation technologies: a review," *Environmental Chemistry Letters*, vol. 19, pp. 797–849, 4 2021. [Online]. Available: <https://link.springer.com/article/10.1007/s10311-020-01133-3>
- [6] L. Zheng, P. Nico, N. Spycher, J. Domen, and A. Credo, "Potential impacts of co₂ leakage on groundwater quality of overlying aquifer at geological carbon sequestration sites: A review and a proposed assessment procedure," *Greenhouse Gases: Science and Technology*, vol. 11, pp. 1134–1166, 10 2021. [Online]. Available: <https://scijournals.onlinelibrary.wiley.com/doi/10.1002/ghg.2104>
- [7] K. Mortezaei, A. Amirlatifi, E. Ghazanfari, and F. Vahedifard, "Potential co₂ leakage from geological storage sites: advances and challenges," *Environmental Geotechnics*, vol. 8, pp. 3–27, 2 2021. [Online]. Available: <https://dx.doi.org/10.1680/jenge.18.00041>
- [8] E. Aursand, S. Dumoulin, M. Hammer, H. I. Lange, A. Morin, S. T. Munkejord, and H. O. Nordhagen, "Fracture propagation control in CO₂ pipelines: Validation of a coupled fluid–structure model," *Engineering Structures*, vol. 123, pp. 192–212, sep 2016. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0141029616302061>
- [9] A. Cosham Atkins Boreas, N. upon Tyne, and R. J. Eiber Robert J, "Fracture Control in Carbon Dioxide Pipelines: The Effect of Impurities," *Proceedings of the Biennial International Pipeline Conference, IPC*, vol. 3, pp. 229–240, jun 2009. [Online]. Available: <https://dx.doi.org/10.1115/IPC2008-64346>
- [10] *Potential impact of CCS leakage on marine communities*, 2014.
- [11] L. H. H. P. Pham, R. Rusli, and L. K. Keong, "Consequence Study of CO₂ Leakage from Ocean Storage," *Procedia Engineering*, vol. 148, pp. 1081–1088, 2016. [Online]. Available: <http://dx.doi.org/10.1016/j.proeng.2016.06.597>
- [12] H. Zhang, S. Zhang, Y. Wang, Y. Liu, Y. Yang, T. Zhou, and H. Bian, "Subsea pipeline leak inspection by autonomous underwater vehicle," *Applied Ocean Research*, vol. 107, p. 102321, feb 2021. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0141118720306362>
- [13] F. Qasem, T. B. Susilo, S. Said, Z. Alarbash, M. Hasan, B. Jabakhanji, T. Beyrouthy, and S. alkork, "Preliminary engineering implementation on multisensory underwater remotely operated vehicle (rov) for oil spills surveillance," in *2019 3rd International Conference on Bio-engineering for Smart Technologies (BioSMART)*, 2019, pp. 1–5.
- [14] S. Hu, A. Feng, J. Shi, J. Li, F. Khan, H. Zhu, J. Chen, and G. Chen, "Underwater gas leak detection using an autonomous underwater vehicle (robotic fish)," *Process Safety and Environmental Protection*, vol. 167, pp. 89–96, nov 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S0957582022007662>

- [15] A. Shukla and H. Karki, "Application of robotics in offshore oil and gas industry— A review Part II," *Robotics and Autonomous Systems*, vol. 75, pp. 508–524, 2016. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0921889015002018>
- [16] *Autonomous Inspection of Subsea Facilities-Gulf of Mexico Trials*, ser. OTC Offshore Technology Conference, vol. Offshore Technology Conference, 04 2012. [Online]. Available: <https://doi.org/10.4043/23512-MS>
- [17] S. Patel, F. Abdellatif, M. Alsheikh, H. Trigui, A. Outa, A. Amer, M. Sarraj, A. Al Brahim, Y. Alnumay, A. Felemban, A. Alrasheed, A. Halawani, H. Jifri, H. Jaleel, and J. Shamma, "Multi-robot system for inspection of underwater pipelines in shallow waters," *International Journal of Intelligent Robotics and Applications*, vol. 8, no. 1, pp. 14–38, mar 2024. [Online]. Available: <https://link.springer.com/article/10.1007/s41315-023-00309-8>
- [18] G. Polverino, P. Phamduy, and M. Porfiri, "Fish and Robots Swimming Together in a Water Tunnel: Robot Color and Tail-Beat Frequency Influence Fish Behavior," *PLoS ONE*, vol. 8, no. 10, pp. 47–50, 2013.
- [19] S. Marras and M. Porfiri, "Fish and robots swimming together: Attraction towards the robot demands biomimetic locomotion," *Journal of the Royal Society Interface*, vol. 9, no. 73, pp. 1856–1868, 2012.
- [20] M. Lighthill, "Hydromechanics of aquatic animal propulsion," *Annual review of fluid mechanics*, vol. 1, no. 1, pp. 413–446, 1969.
- [21] M. U. Masood and Z. Chen, "Visual servoing control of robotic fish for underwater pipeline tracking," *International Journal of Intelligent Robotics and Applications*, pp. 1–28, 2025.
- [22] W. Zuo, F. Fish, and Z. Chen, "Bio-inspired design, modeling, and control of robotic fish propelled by a double-slider-crank mechanism driven tail," *Journal of Dynamic Systems, Measurement and Control, Transactions of the ASME*, vol. 143, pp. 1–10, 2021.
- [23] M. U. Masood, T. Kaaya, and Z. Chen, "Constrained model predictive control of variable buoyancy device," *IEEE/ASME International Conference on Advanced Intelligent Mechatronics, AIM*, vol. 2023-June, pp. 936–941, 2023.